

MCRec: Few-Shot Multimodal Cover Recommendation via User Interest Profiles

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Abstract

Recommendation systems play a central role in modern services, yet often treat item cover images as static attributes, overlooking their influence on user decisions. We introduce the task of cover recommendation and study few-shot, interaction-free selection using multimodal user interest profiles. To address cold-start and sparsity challenges in traditional methods, we propose Multimodal Cover Recommendation (MCRec), a framework that leverages Vision-Language Models (VLMs) for multimodal feature extraction. Our approach includes: (1) a Text-Guided Visual Interest Aggregation network (TGVIA) integrating visual and textual representations; (2) multimodal interest embeddings fused via templated prompts; and (3) a multimodal-driven textual inversion technique enabling training-free generalization to new scenarios. We further propose MCRec+, a fine-tuning variant using hybrid sampling. To support evaluation, we construct three benchmarks and propose two new metrics. Extensive experiments show our methods significantly outperform baselines across datasets, especially with average gains of 3.72% in Recall@1, 1.70% in APMS and 1.25% in MPMS on MCRec. Code and data are publicly available from <https://github.com/WeixinZhengRec/MCRec>.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Cover Recommendation, Multimodal, VLMs, Few-shot learning

ACM Reference Format:

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1 Introduction

Recommendation systems are vital for mitigating information overload in modern information ecosystems. In typical applications

such as e-commerce[22, 31] and short-video platforms[5, 46], the primary task of traditional recommendation systems is to predict user preferences for items—essentially answering “what item should be recommended to the user?”. These systems generally assume that the visual representation of an item (e.g., a product’s main image or video covers) is static and unique[13, 23, 24, 28], overlooking the fact that different visual presentations of the same item may strongly influence its attractiveness to users[12].

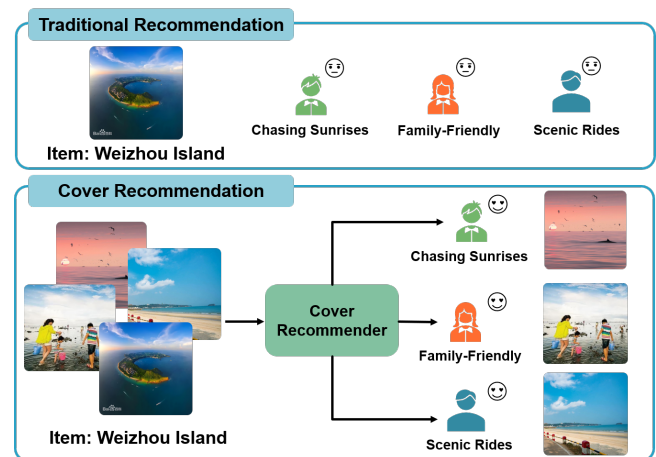


Figure 1: The difference between traditional recommendation and cover recommendation.

To address this gap, we introduce the **cover recommendation** task, which focuses on answering “how should an item be visually presented to a specific user to maximize its appeal?”. The goal of cover recommendation is to capture personalized visual preferences and dynamically select, for each item, the cover most likely to trigger user interaction. This enhances exposure, increases click-through[18] and provides significant commercial value[43]. For instance, e-commerce platforms may display fashion street photos of a product to users who favor stylish looks, and fabric details to users who value practicality, thereby boosting clicks and purchases. Similarly, streaming services can promote a single movie using posters tailored to each user, featuring their favored actors to boost engagement and retention. The difference between traditional recommendation and cover recommendation is shown in Figure 1.

However, cover recommendation faces three key challenges: cold-start issues, data sparsity, and attribution difficulty. First, cover recommendation is a typical multi-faceted cold-start scenario. In

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practice, beyond the well-known cold-start users[29, 49] and items [33, 44] in traditional systems, there is also the cold-start cover issue. For instance, e-commerce platforms may design new eye-catching promotional images or adjust product cover backgrounds [12]. These new covers lack historical interaction data at launch, making it hard to reliably predict appeal via past interaction signals.

Second, the data sparsity problem is more severe than item recommendation. While traditional recommendation [30, 45, 48] relies on user-item binary interactions (e.g., clicks, purchases), cover recommendation requires granular user-item-cover ternary relationships. Since a user typically interacts with an item only once, and each exposure shows only one cover, the available user-item-cover-level data is extremely sparse.

Third, accurately attributing interaction behavior is challenging [17], leading to significant noise in supervision signals. User behavior is driven by multiple factors, including the item's content (e.g., movie plot, theme, actors), visual presentation of the cover, and other contextual conditions[17]. It is hard to disentangle whether a click stems from the cover, the content, or other influences. Moreover, users typically have only one exposure to each item and see just one cover version, precluding direct modeling of different covers' true impacts. These issues severely hinder the construction of reliable supervision signals that reflect the covers' visual appeal based on interactions. Consequently, models may learn spurious correlations related to item content, compromising the accuracy and generalization performance of cover recommendations.

Traditional collaborative filtering[40] approaches and their variants (e.g., NCF[21], LightGCN[20]) rely on user-item interaction matrices, making them poorly suited for this scenario. We turn to mining user preferences directly from cover content, with a particular emphasis on few-shot learning scenarios. Recent advances in vision-language models (VLMs) provide a promising alternative. Trained on large-scale image-text pairs, VLMs possess strong cross-modal alignment and representation abilities[27, 34]. In particular, textual inversion enables the embedding of new visual concepts into the text space of a pre-trained VLM using only a few examples[1, 2, 15, 37]. This allows to establish explicit associations between user interest labels (such as cyberpunk and minimalist, which are often provided during user registration) and cover styles, and constructs an "interest-cover" bridge to lessen dependence on historical interactions and resolve attribution ambiguity.

Building on this, we propose a new paradigm for cover recommendation that does not depend on user historical interactions: Few-shot **M**ultimodal **C**over **R**ecommendation via Semantic User Profiles (MCR_{ec}). MCR_{ec} uses user interest labels with a small set of covers, and leverages pre-trained VLMs and textual inversion to extract and fuse multimodal interest features. This produces compact, semantically rich user interest representations that enable effective recommendations even in full cold-start scenarios. Specifically, we design a Text-Guided Visual Interest Aggregation (TG_{VIA}) module to fuse multi-level visual interest features and map them to the semantic space. This history-free pipeline directly tackles sparsity and attribution issues. By leveraging the generalization power of pre-trained VLMs, it can recommend for new users, interests, or covers without training, thus mitigating the cold start problem.

Additionally, cover recommendation lacks standardized benchmarks. We filled this gap by constructing three real-world datasets

based on user-image interactions and interest labels (pinterest-small[16], pinterest-large[16], reasoner[8]). We also introduce dedicated evaluation metrics that jointly assess visual attractiveness, interest relevance, and hit rate, enabling comprehensive evaluation.

The main contributions of this work are as follows:

- We propose MCR_{ec} for the new task of cover recommendation: a VLM-based method integrating textual inversion and TG_{VIA} for multimodal representation, with unsupervised pre-training enabling generalization to new scenarios. We also contribute three benchmark datasets and two metrics.
- We further propose MCR_{ec}+, a vector fine-tuning method based on proposed difficulty-aware hybrid sampling. This improves the model's ability to distinguish visually similar covers, enhancing fine-grained visual preference modeling.
- Experiments on the three datasets show that MCR_{ec} significantly outperforms baseline models in key metrics, while MCR_{ec}++ further improves performance and achieves SOTA.

2 Related Work

2.1 Vision-language models

VLMs, such as CLIP [34] and BLIP[27], are pre-trained on large-scale image-text pairs to map image and textual modalities into a unified semantic space, thereby achieving strong feature extraction capability, modality alignment performance, and generalization[36]. These capabilities have enabled the widespread application in various areas, including zero-shot image classification[39], text-image retrieval[6], and image generation[47]. Furthermore, many studies directly employ CLIP's text and image encoders as feature extractors [25], further validating the powerful potential of VLMs in cover recommendation. However, CLIP has inherent limitations in representing personalized concepts, struggling to adequately capture the specific preferences. Therefore, to effectively apply CLIP to cover recommendation task, it is essential to equip it with personalized modeling capabilities. To this end, we propose a novel approach to adapt CLIP for the downstream task of cover recommendation.

2.2 Personalized Image Retrieval based on Textual Inversion

To enable few-shot personalization in CLIP, existing research has primarily explored textual inversion techniques that adapt the model using a small set of specific images[37]. The core idea is to map images into a pseudo-token to represent personalization concepts. For instance, PALAVRA[10] employs a two-stage personalization framework. It first uses textual inversion on a large-scale dataset to learn a general pseudo-token from image sets, then fine-tunes it for specific tasks. However, the second stage is computationally expensive.

Another train-free approach uses textual inversion with an MLP network to map single images into pseudo-tokens for one-shot retrieval. For example, Pic2Word[38] aligns images and text in CLIP space, SEARLE[2] employs unsupervised inversion and distillation, and LinCIR[19] uniquely trains its projection module using only language data.

However, existing methods are ill-suited for cover recommendation, as they process only single images without textual guidance.

In contrast, our novel solution maps a set of interest cover images to a pseudo-token under textual guidance and adopts a fine-tuning strategy to enhance cover recommendation performance.

3 Preliminaries

This section first formally defines the cover recommendation problem and presents the key notations used. Then we introduce the foundational knowledge on which our method is based.

3.1 Problem Formulation

Given a user u and an item i , the user's interest label set is $\mathcal{T}_u = \{t_k \mid k = 1, \dots, K_u\}$. The item's cover set is $C_i = \{c_k \mid k = 0, \dots, K_i\}$. The covers are the potential main images used to visually present the item. Specifically, we designate c_0 as the target interest cover of user. Each interest $t \in \mathcal{T}_u$ is characterized by its textual label $label_t$ and a small set of interest covers $C_t = \{c_k \mid k = 0 \dots K_t\}$, which jointly capture interest semantics. Consequently, the task of the cover recommendation is to learn a function f that predicts the target cover based on the user's interests and the item's candidate covers:

$$\hat{c} = f(\mathcal{T}_u, C_i; \theta) \quad (1)$$

Here, f is the recommendation model to be learned, and θ denotes its parameters. Table 1 lists the main notations used in this paper.

Table 1: Main notations used in this paper

Symbol	Description
$\mathcal{U}, \mathcal{I}, \mathcal{T}$	Sets of users, items, and interest labels
u, i, t	User/item/interest label indices
C_i, C_t	Set of candidate cover set of item i , interest cover of label t
$c_0, \hat{c}_k$	Real preferred cover $c_0 \in C_i$ and k -th recommended cover of user u under interaction with item i
$\mathcal{T}_u, \mathcal{T}_c$	Set of real interest label of user u , interest label set of cover c
t_u, t_c	One interest label of user u and item cover c
e_c, e_t, e_u	Cover embedding, multimodal interest embedding, and user embedding
w_t	visual interest pseudo-token
S, S^*	A sentence that is related to a special label, and a sentence that replaces the label in S with the visual interest pseudo-token.
e_S, e_{S^*}	Embedding of sentence S and S^*
K_u, K_t, K_i	The number of users, item covers, and items
f_T, f_I	Text encoder, and image encoder

3.2 Contrastive Language-Image Pre-training

Contrastive Language-Image Pre-training (CLIP) [34] is a pre-trained VLM that maps images and their corresponding text into a shared embedding space via contrastive learning on large-scale image-text data. It provides strong cross-modal understanding and retrieval capabilities. CLIP comprises an image encoder f_I and a text encoder f_T . In this work, we use CLIP to obtain representations for multimodal user interests and item covers. Because CLIP is trained to

place semantically similar concepts close together across modalities, we can perform personalized cover recommendations by measuring the similarity between a user and a cover by their embeddings.

3.3 Textual Inversion: Projection to Token Embeddings in VLMs

Textual inversion is a technique for mapping image embeddings into the token embedding space[15]. By adding a mapping network and applying textual inversion, few-shot images can be used to represent a new visual concept as a pseudo-token in the VLM's token embedding space[2, 10, 38]. This achieves semantically representing personalized visual interests.

During pre-training, S^* is constructed by replacing the label in the label sentence S with a learned pseudo-token. Here, S is the sentence associated with the label. It is constructed using a template like "A photo of {label}"[38], or generated by a Large Language Model[2]. Subsequently, the text encoder encodes the sentences S, S^* into e_S , and e_{S^*} , respectively. Then, the mapping network is trained to align visual content into the text embedding space. Textual inversion based on text alignment by cosine similarity[2] can be expressed as:

$$\mathcal{L}_{\text{text}} = 1 - \cos(e_S, e_{S^*}) \quad (2)$$

Textual inversion based on image alignment by SimCLR[7] can be written as:

$$\mathcal{L}_{\text{img}} = \text{SimCLR}(e_c, e_{S^*}) \quad (3)$$

where e_c denotes image embedding. In our method, we combine both text-based and image-based alignment for textual inversion.

4 Methodology

In this section, we present MCRRec, a CLIP-based framework for cover recommendation. It first introduces the TGVIA module to aggregate visual interests under textual guidance and fuse them with textual interests, yielding integrated representations. A multimodal-driven textual inversion pre-training is employed to enhance generalizability. Building upon this, we further propose MCRRec+, which fine-tunes the interest representations via a hybrid sampling strategy. During inference, the user embedding is obtained by average pooling over all user interest representations, and recommendations are generated by calculating its similarity with candidate cover embeddings using CLIP. Our overall architecture is shown in Figure 2, with core components detailed in subsequent subsections.

4.1 MCRRec: Few-Shot Multimodal Cover Recommendation

MCRRec extracts user interest information from both visual and textual modalities. We design TGVIA to obtain visual features highly aligned with the interest by multimodal-driven textual inversion.

4.1.1 Text-Guided Visual Interest Aggregation. TGVIA comprises three core modules: Context-Aware Intra-Modal Attention, Text-Aware Cross-Modal Attention, and Multi-Level Feature Fusion. It first processes visual inputs through the first two modules to generate context-aware and text-aware visual features. The Fusion Module then integrates these with the original visual features to produce the final visual interest pseudo-token.

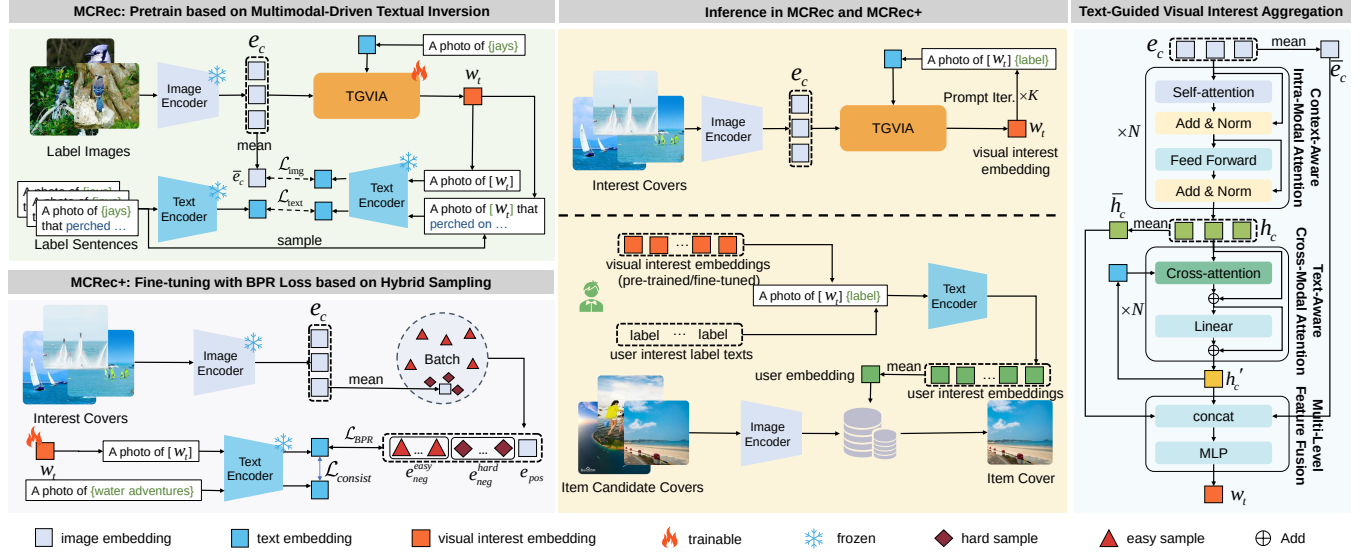


Figure 2: Overview of our MCRRec and MCRRec+

Context-Aware Intra-Modal Attention. This module employs a self-attention mechanism to model contextual associations among input interest images (i.e., covers with the same interest label), aiming to mine their intrinsic connections and contextual information. It provides semantically rich visual representations for subsequent cross-modal interaction and hierarchical feature fusion. We first encode the interest cover set C_t using the CLIP image encoder to obtain an embedding matrix $\mathcal{E} = \{e_c | c \in C_t\}$. This matrix serves as the input for computing the query, key, and value matrices of the attention mechanism[42]:

$$Q = f_Q(\mathcal{E}), \quad K = f_K(\mathcal{E}), \quad V = f_V(\mathcal{E}) \quad (4)$$

where f_Q, f_K, f_V are learnable linear projection layers.

Next, a multi-head self-attention layer is used to construct context-aware visual features for each cover. For cover c , its output is:

$$\text{MHA}_c = \text{Norm}(f_{\text{out}} \left(\left[\text{softmax} \left(\frac{q_c K^T}{\sqrt{d_q}} \right) V \right]_{j=1}^{k_h} \right) + e_c) \quad (5)$$

where $\text{Norm}(\cdot)$ denotes layer normalization, q_c is the query vector in Q corresponding to cover c , d_q is the dimension of the q_c , k_h is the number of attention heads, and f_{out} is the output projection layer. Finally, the context-aware visual feature h_c is obtained through a Feed-Forward Network (FFN):

$$h_c = \text{Norm}(\text{FFN}(\text{MHA}_c) + \text{MHA}_c) \quad (6)$$

where $\text{Norm}(\cdot)$ denotes layer normalization. This structure dynamically captures information interactions between different covers, enhancing the interest representation capability of the images. Context-Aware Intra-Modal Attention (CAIMA) can be stacked to capture richer context:

$$\mathcal{H}^i = \text{CAIMA}_i(\mathcal{H}^{i-1}), \quad i = 1, \dots, L \quad (7)$$

where $\mathcal{H}^i = \{h_c^i | c \in C_t\}$ denotes the context features at the i -th layer, and L is the total number of stacked CAIMA layer.

Text-Aware Cross-Modal Attention. To suppress background noise unrelated to the interest in the covers, we introduce a text-guided cross-modal attention mechanism. After obtaining context-aware visual features, we use textual information to guide and calibrate them. First, we insert the user interest label text of t into a predefined template sentence S (e.g., “A photo of {label}”) and encode it by the CLIP text encoder to obtain its embedding e_s . Then, multi-head cross-attention is computed, using the text embedding e_s as the query $q'_s = f'_Q(e_s)$, and the context-aware visual features of covers $\mathcal{H} = \{h_c | c \in C_t\}$ as the key $K' = f'_K(\mathcal{H})$ and value $V' = f'_V(\mathcal{H})$. The output of the multi-head cross-attention layer is passed through a linear transformation to obtain the text-aware visual feature h' . We use residual connections to enhance feature expressiveness like Equation 5 and Equation 6:

$$\text{MHA}'_s = \text{Norm}(f_{\text{out}} \left(\left[\text{softmax} \left(\frac{q'_s K'^T}{\sqrt{d'_q}} \right) V' \right]_{j=1}^{k_h} \right) + e_s) \quad (8)$$

$$h' = \text{Linear}(\text{MHA}'_s) + \text{MHA}'_s \quad (9)$$

where $\text{Linear}(\cdot)$ denotes a learnable linear projection layer. This module can also be stacked to further improve representation quality like Equation 7.

Multi-Level Feature Fusion. Background information present in the original visual features can provide cues for interest expression. Therefore, after obtaining the context-aware feature h_c and text-aware feature h' , we fuse them with the original cover image embedding e_c to retain the background cueing ability from the original visual information. The features are fused and projected into the CLIP text space to generate the visual interest pseudo-token

w_t , calculated as:

$$\begin{cases} w_t = \text{MLP}([\bar{e}_c; \bar{h}_c; h']) \\ \bar{e}_c = \frac{1}{K_t} \sum_{c \in C_t} e_c \\ \bar{h}_c = \frac{1}{K_t} \sum_{c \in C_t} h_c \end{cases} \quad (10)$$

where K_t presents the number of interest covers, $[\cdot]$ denotes vector concatenation, and MLP is a multilayer perceptron.

TGVIA generates a rich and aligned visual interest representation by aggregating multi-level visual features, thereby supporting high-quality personalized recommendations. Subsequently, we fuse the visual interest pseudo-token w_t with the interest text using the prompt template to form a multimodal interest representation e_t . The prompt enhances the model's generalization, and multimodal fusion further improves the representation effectiveness. Since the template sentence directly affects the text-guided effect, we adopt prompt iteration during inference to improve the representation quality. Initially, S is constructed using only the interest label text. This initial representation leverages the textual prompt ("A photo of $[w_t]$ $\{label\}$ ") to preliminarily extract a visual representation, which in turn refines S . Appendix A.3.3 analyzes the impact of iteration times on balancing performance and computational cost.

4.1.2 Multimodal-Driven Textual Inversion. To achieve better performance in the few-shot scenario, we propose multimodal-driven textual inversion. This method combines the advantages of textual inversion in few-shot learning with multimodal information to learn a generalized TGVIA.

During pre-training, we construct triplets of "label, image, label sentence" (t, c, S), where the image acts as a cover to learn visual feature representations. We use COCO 2017¹ as the image repository and Open Images V7² as the label repository. First, CLIP is used to compute image-text similarity, and k images are assigned to each label. Here, a larger k increases diversity but may reduce image-text alignment quality. We determine its optimal value experimentally (see Appendix A.3.2). Then, we use ChatGLM4³ to generate richer semantic descriptions for each image's label via the prompt: "A photo of $\{label\}$ that". This results in triplet samples (t, c, S).

During training, for each label in every batch, we randomly sample K_t triplet samples to construct the training image set C_t . This random combination enhances sample diversity, benefiting generalization. TGVIA is then used to extract the visual interest pseudo-token w_t . One label sentence S is selected, and w_t is embedded into it by replacing the label, resulting in S^* . Finally, we use the cosine similarity in the textual modality to minimize the semantic distance between S and S^* , defining the text alignment loss:

$$\mathcal{L}_{\text{text}} = 1 - \cos(e_S, e_{S^*}) \quad (11)$$

where e_S and e_{S^*} are the embedding vectors of S and S^* obtained via the CLIP text encoder, respectively.

Since the label sentences S generated by the LLM based solely on the label text can not precisely describe the image content and

contain noise, we use a cleaner, fixed template sentence (e.g., "A photo of $\{label\}$ ") to construct the target textual representation S' . The visual alignment objective is to align the embedding $e_{S'^*}$ with the visual representation $\bar{e}_c$ of the sampled image set C_t . We employ SimCLR[7] loss:

$$\mathcal{L}_{\text{img}} = \frac{1}{|B|} \sum_{i \in B} \left(-\log \frac{e^{\cos(\bar{e}_c, e_{S'^*})/\tau}}{\sum_{j \in B} e^{\cos(\bar{e}_c, e_{S'^*})/\tau} + \sum_{j \neq i} e^{\cos(e_{S'^*}, e_{S'^*})/\tau}} - \log \frac{e^{\cos(\bar{e}_c, e_{S'^*})/\tau}}{\sum_{j \in B} e^{\cos(e_{S'^*}, \bar{e}_c)/\tau} + \sum_{j \neq i} e^{\cos(\bar{e}_c, \bar{e}_c)/\tau}} \right) \quad (12)$$

where $\bar{e}_c$ is obtained by average pooling the CLIP embeddings of the image set, $e_{S'^*}$ is the predicted visual embedding, and τ is the temperature coefficient.

The final multimodal-driven textual inversion loss function jointly combines the text alignment and image alignment losses:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{text}} + \lambda_2 \mathcal{L}_{\text{img}} \quad (13)$$

where λ_1 and λ_2 are adaptive weights, scaled by the inverse of the corresponding loss values for $\mathcal{L}_{\text{text}}$ and $\mathcal{L}_{\text{img}}$. This stabilizes the multi-loss optimization for multimodal training, significantly improving the accuracy and generalization of interest representations in few-shot learning.

4.2 MCRRec+: Fine-tuning with BPR Loss based on Hybrid Sampling

To further enhance performance on cover recommendation under a new interest domain $\mathcal{T}$, we propose MCRRec+, a lightweight fine-tuning method based on the pre-trained MCRRec. MCRRec+ directly fine-tunes the visual interest embedding generated by MCRRec. We use the Bayesian Personalized Ranking (BPR)[35] loss as the optimization objective. To maintain stability during fine-tuning and enable effective learning, we design a hybrid sampling strategy.

4.2.1 Hybrid Sampling Strategy. For an interest label of t , the average of the image embeddings from the interest cover set is used as the positive sample visual supervision signal:

$$e_{\text{pos}} = \frac{1}{K_t} \sum_{c \in C_t} e_c \quad (14)$$

For each positive sample e_{pos} , we identify potential negative samples within the same batch. To mitigate the false negative problem, we only select samples corresponding to labels whose text similarity with the interest label t is below 90%. Considering the high similarity between candidate covers for the same item in the cover recommendation task, we propose a hybrid negative sampling strategy that introduces hard negative samples to enhance the model's discrimination ability. Specifically, we select the most similar to the positive sample as the hard negative sample $e_{\text{neg}}^{\text{hard}}$ based on CLIP visual similarity. And, easy negative samples $e_{\text{neg}}^{\text{easy}}$ are then randomly sampled from the remaining negatives. Each positive sample corresponds to ten sampled negative samples, with 20% of them being hard negatives.

¹<https://cocodataset.org/>

²<https://storage.googleapis.com/openimages/web/index.html>

³<https://open.bigmodel.cn/>

4.2.2 Bayesian Personalized Ranking Loss based on Hybrid Sampling. Based on the positive and sampled negative samples, we finetune the visual interest pseudo-token by the loss like BPR[5]:

$$\mathcal{L}_{\text{BPR}} = -\frac{1}{|B|} \sum_{(t, \text{pos}, \text{neg}) \in B} \ln \sigma(\hat{y}_{t, \text{pos}} - \hat{y}_{t, \text{neg}}) + m \quad (15)$$

where B represents the samples in the batch, $\sigma(\cdot)$ is the sigmoid function, and $\hat{y}_{t, \text{pos}}$ and $\hat{y}_{t, \text{neg}}$ are the model's prediction scores between the interest representation $\hat{e}_t$ and the positive sample e_{pos} and the negative sample e_{neg} , respectively. The m term forces the model to learn distinctions between the embeddings.

To prevent the fine-tuned model from overfitting to the noise of covers, which could lead to a drift in the interest features, we introduce an interest text consistency constraint loss. This loss directly minimizes the Mean Squared Error between the interest text embedding e_t and the fine-tuned interest embedding $\hat{e}_t$:

$$\mathcal{L}_{\text{consist}} = \frac{1}{|B|} \sum_{t \in \mathcal{T}_B} \|\hat{e}_t - e_t\|_2^2 \quad (16)$$

where $\mathcal{T}_B$ denotes the set of labels contained in batch B .

The final loss function for MCRc+ combines $\mathcal{L}_{\text{BPR}}$ and $\mathcal{L}_{\text{consist}}$:

$$\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{\text{BPR}} + \lambda \cdot \mathcal{L}_{\text{consist}} \quad (17)$$

where λ is a hyperparameter weighting the strength of the consistency constraint. By minimizing $\mathcal{L}_{\text{Hybrid}}$, MCRc+ learns fine-grained, more robust, and better-adapted interest visual representations for the specific target scenario, thereby obtaining superior user embeddings and improving cover recommendation performance.

5 Experiments

We conduct extensive experiments using three constructed benchmark datasets, as detailed in Appendix A.1: PC-S, PC-L, and RC.

5.1 Experiment Setup

5.1.1 Implementation Details. Our work is implemented in PyTorch and runs on a server with NVIDIA GeForce RTX3090. We utilize the CLIP ViT-L/14⁴ model as VLMs. The number of interest covers K_t are set to 3 to represent the few-shot scenario. The numbers of self-attention and cross-attention layers are set to 2 and 3, respectively, with a dropout rate of 0.2. The AdamW[32] is employed for training with 0.1 weight decay. The batch size is 128, and the hidden layer dimension is 1024.

During pre-training, each label from the training vocabulary dictionary (Open Images V7) is associated with the top 2 most similar images from the predefined image gallery (COCO 2017). This stage uses a contrastive temperature of 0.08 and a learning rate of 5×10^{-4} for 20 epochs. When fine-tuning, the parameters are trained for 5 epochs with a learning rate of 0.1.

⁴<https://huggingface.co/openai/clip-vit-large-patch14>

5.1.2 Evaluation Protocols. The cover recommendation task aims to improve the accuracy of recommended covers, with emphasis on the quality of the TOP-1 result. We measure recommendation accuracy from two complementary dimensions: interest label semantic matching and actual user interaction behavior.

First, based on interest label semantic matching, we propose two new metrics: Average Preference Matching Score (APMS) and Maximum Preference Matching Score (MPMS). These quantify the semantic alignment between user interest and recommended covers. For each interaction between user $u \in \mathcal{U}$ and item $i \in \mathcal{I}$, the associated interest label set is $\mathcal{T}_u$. User u 's preference centroid vector is defined as $\mathbf{g}_u = \frac{1}{|\mathcal{T}_u|} \sum_{t \in \mathcal{T}_u} \text{encode}(t)$, where $|\mathcal{T}_u|$ is the number of user interest labels, and $\text{encode}(\cdot)$ is a text encoding function mapping label texts to semantic vectors. We implement this encoding using the pre-trained all-MiniLM-L6-v2⁵ that effectively converts text to semantic vectors for similarity computation. Each label $\hat{i} \in \hat{\mathcal{T}}_c$ of a recommended cover $\hat{c}$ is similarly encoded as $\hat{\mathbf{g}}_i = \text{encode}(\hat{i})$. The Preference Matching Score (PMS) between user u and cover label $\hat{i}$ is their cosine similarity:

$$\text{PMS}(u, \hat{i}) = \cos(\hat{\mathbf{g}}_i, \mathbf{g}_u) \quad (18)$$

Based on this, MPMS and APMS are formalized as:

$$\text{MPMS} = \frac{1}{|\mathcal{D}|} \sum_{(u, i, c) \in \mathcal{D}} \max_{\hat{i} \in \hat{\mathcal{T}}_c} \text{PMS}(u, \hat{i}) \quad (19)$$

$$\text{APMS} = \frac{1}{|\mathcal{D}|} \sum_{(u, i, c) \in \mathcal{D}} \frac{1}{|\hat{\mathcal{T}}_c|} \sum_{\hat{i} \in \hat{\mathcal{T}}_c} \text{PMS}(u, \hat{i}) \quad (20)$$

where $\mathcal{D}$ is the set of all user-item-cover interaction triplets. MPMS reflects the maximum alignment between user preferences and the most relevant cover label, measuring the cover's ability to satisfy core user interest. APMS reflects the average alignment across all cover labels, measuring the cover's overall appeal to the user.

Second, based on actual user interaction behavior, we introduce Recall@1 to directly measure the accuracy of correctly recommending the target cover among candidates. While TOP-1 recommendation is our primary metric, we also evaluate overall ranking quality using mean Reciprocal Rank (mRR)[11], Recall@K[2], NDCG@K[26], and mean Average Precision (mAP@K)[3] for a comprehensive assessment. However, as the task prioritizes high-quality top results, larger K values are less indicative of core performance. Consequently, we set K to 2 and 3.

5.1.3 Baselines. We compare our proposed MCRc and MCRc+ method against the CLIP variants: CLIP-text, CLIP-image, and CLIP-multimodal. Personalized image retrieval: PALAVRA. Zero-shot composed image retrieval: Freedom, OSrCIR, Pic2Word, SEARLE, and LinCIR. To ensure a fair comparison, all methods are based on the same CLIP ViT-L/14 backbone and adopt a unified input preprocessing pipeline. More details are provided in Appendix A.4.

5.2 Overall Performance

Table 2 presents the performance comparison between our proposed method and various baseline models across the PC-L, PC-S, and RC datasets. On the PC-S dataset, label semantics are more specific, the distinction between labels is high, and the visual differences

⁵<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

Table 2: The overall experiment performance on three datasets. The bold is the best result and the underline is the second-best

Dataset	Method	MPMS	APMS	mRR	Recall@1	Recall@2	NDCG@2	mAP@2	Recall@3	NDCG@3	mAP@3
PC-S	CLIP-text	0.6002	0.5289	0.6655	0.5122	0.6748	0.6148	0.5935	0.7584	0.6566	0.6214
	CLIP-image	0.8459	0.7988	0.8608	0.7862	0.8708	0.8396	0.8285	0.9276	0.8680	0.8474
	CLIP-multimodal	0.8664	0.8144	0.8799	0.8185	0.8842	0.8599	0.8513	0.9321	0.8839	0.8673
	Freedom	0.7559	0.6911	0.7956	0.6882	0.8263	0.7753	0.7572	0.8764	0.8004	0.7739
	OSrCIR	0.6280	0.5650	0.6860	0.5145	0.7539	0.6655	0.6342	0.8229	0.7001	0.6572
	Pic2Word	0.8927	0.8316	0.9155	0.8508	0.9555	<u>0.9168</u>	0.9031	<u>0.9788</u>	0.9285	0.9109
	SEARLE	<u>0.9158</u>	<u>0.8526</u>	<u>0.9261</u>	<u>0.8820</u>	<u>0.9343</u>	0.9150	<u>0.9081</u>	0.9744	<u>0.9350</u>	<u>0.9215</u>
	LinCIR	0.8738	0.8091	0.8989	0.8374	0.9165	0.8873	0.8769	0.9532	0.9057	0.8892
	MCRec	0.9404	0.8778	0.9477	0.9143	0.9555	0.9403	0.9349	0.9855	0.9553	0.9449
	PALAVRA	0.9393	0.8785	0.9459	0.9109	0.9555	0.9390	0.9332	0.9866	0.9546	0.9436
	MCRec+	0.9517	0.8845	0.9599	0.9332	0.9710	0.9571	0.9521	0.9866	0.9649	0.9573
PC-L	CLIP-text	0.6653	0.4897	0.5513	0.3865	0.5268	0.4750	0.4566	0.6234	0.5233	0.4888
	CLIP-image	0.6406	0.4876	0.4945	0.3337	0.4550	0.4102	0.3944	0.5385	0.4520	0.4222
	CLIP-multimodal	0.6721	0.5088	0.5414	0.3860	0.5108	0.4647	0.4484	0.5939	0.5063	0.4761
	Freedom	0.6786	0.5059	0.5670	0.4062	0.5431	0.4926	0.4747	0.6395	0.5408	0.5068
	OSrCIR	0.6469	0.4790	0.5331	0.3620	0.5086	0.4545	0.4353	0.6064	0.5034	0.4679
	Pic2Word	<u>0.7189</u>	<u>0.5380</u>	<u>0.6173</u>	<u>0.4665</u>	<u>0.6036</u>	<u>0.5530</u>	<u>0.5351</u>	<u>0.6907</u>	<u>0.5966</u>	<u>0.5641</u>
	SEARLE	0.7097	0.5292	0.6088	0.4510	0.6002	0.5452	0.5256	0.6900	0.5901	0.5556
	LinCIR	0.6988	0.5234	0.5971	0.4387	0.5832	0.5298	0.5109	0.6761	0.5763	0.5419
	MCRec	0.7266	0.5455	0.6267	0.4762	0.6160	0.5644	0.5461	0.7016	0.6072	0.5747
	PALAVRA	0.7108	0.5365	0.6157	0.4610	0.6072	0.5533	0.5341	0.6955	0.5974	0.5636
	MCRec+	0.7351	0.5509	0.6427	0.4949	0.6359	0.5839	0.5655	0.7207	0.6263	0.5938
RC	CLIP-text	0.5907	0.3334	0.3008	0.1198	0.2080	0.1754	0.1639	0.2995	0.2212	0.1944
	CLIP-image	0.5606	0.3222	0.2190	0.0572	0.1155	0.0940	0.0864	0.1769	0.1247	0.1068
	CLIP-multimodal	0.5720	0.3253	0.2297	0.0666	0.1255	0.1037	0.0960	0.1893	0.1357	0.1173
	Freedom	0.5876	0.3328	0.2997	0.1162	0.2062	0.1730	0.1612	0.3022	0.2210	0.1932
	OSrCIR	0.5819	0.3306	0.2961	0.1167	0.2031	0.1712	0.1599	0.2924	0.2158	0.1896
	Pic2Word	0.5902	0.3342	0.3133	0.1264	0.2205	0.1858	0.1735	0.3233	0.2372	0.2077
	SEARLE	<u>0.5956</u>	0.3360	<u>0.3435</u>	0.1510	<u>0.2677</u>	<u>0.2247</u>	<u>0.2094</u>	<u>0.3686</u>	<u>0.2751</u>	<u>0.2430</u>
	LinCIR	0.5982	<u>0.3361</u>	0.3430	<u>0.1531</u>	0.2606	0.2209	0.2068	0.3657	0.2735	0.2419
	MCRec	0.5982	0.3386	0.3617	0.1614	0.2967	0.2468	0.2291	0.4094	0.3031	0.2666
	PALAVRA	0.6004	0.3377	0.3472	0.1553	0.2696	0.2274	0.2125	0.3798	0.2825	0.2492
	MCRec+	0.6174	0.3467	0.3866	0.1914	0.3271	0.2770	0.2592	0.4382	0.3325	0.2963

among candidate covers are significant. As a result, all methods perform well on the dataset. A downward trend in performance is observed for all methods from PC-S to PC-L and to RC, indicating a progressive increase in the challenge posed by the datasets. This is due to the significantly increased difficulty of the cover recommendation task, which stems from users' multiple interest labels and the heightened similarity among candidate covers.

Notably, as the similarity between the target cover and candidate covers increases, the performance of the image-only CLIP-image method degrades more significantly than the text-only CLIP-text method. On the PC-L and RC datasets, the performance of CLIP-image even falls below CLIP-text. This degradation is particularly pronounced on the RC dataset, where CLIP-image's performance on Recall@1 is worse than random recommendation (Recall@1 of random recommendation is 0.1). This reflects the intra-class characteristics among different covers of the same item. Generally, the

multimodal CLIP-multimodal method outperforms CLIP methods using only single-modality text or image features, demonstrating the effectiveness of multimodality. However, on the RC dataset, the performance of the multimodal fusion method declines. This may be attributed to the excessively high similarity in the image features among the candidate elements, which makes it difficult for simple feature extraction and fusion strategies to capture discriminative cues effectively.

We observe that both Freedom and OSrCIR, which utilize discrete explicit text personalizations, perform poorly. Freedom introduces external images and labels, leading to a significant loss of personalized user characteristics. For OSrCIR, we argue it works effectively only when prompts align with the specific task, the LLM, and CLIP weights. However, cover recommendation requires more complex adaptation and is theoretically limited in fully conveying personalized visual information through text alone. Furthermore,

we find that OSrCIR introduces substantial noisy content unrelated to personalized visual features. In the comparison of few-shot methods, other models exhibit their respective advantages on different datasets. However, our method demonstrates a significant advantage across all evaluation metrics, achieving state-of-the-art performance. On the key metrics of Recall@1, APMS, and MPMS, our method improves upon the second best by an average of 3.72%, 1.70%, and 1.25%, respectively. In the overall ranking metric mRR, MCRRec shows an average improvement of 3.05%. It consistently demonstrates an advantage in both key and various supplementary ranking metrics. These gains show our method effectively extracts and fuses discriminative features from interest label texts and images to form robust user preference representations. Remarkably, pre-trained MCRRec still outperforms the fine-tuned PALAVRA model on most datasets, supporting the effectiveness.

When fine-tuned, MCRRec+ achieves a significant performance advantage over PALAVRA. Crucially, our method improved substantially after fine-tuning, with an average improvement of 1.86% on MPMS, 1.37% on APMS, 8.20% on Recall@1, and 3.57% on mRR. We attribute this to two factors: its adaptive cover recommendations by BPR Loss based on the Hybrid Sample and the consistency of the data distribution. These characteristics show that our method can adapt to different interest labels in cover recommendations.

5.3 Ablation Study

This section presents ablation studies on MCRRec and MCRRec+ across three datasets to evaluate the contribution of each component. To this end, we analyze the two model stages separately to avoid confounding effects. First, we ablate key pre-training components in MCRRec. Then, for MCRRec+, we use the pre-trained embeddings as initial features to assess the impact of its subsequent components, as detailed below:

MCRRec

- w/o TGVIA: Replaces the TGVIA with a 3-layer MLP to directly extract common features from the image set, without text-aware aggregation.
- w/o $\mathcal{L}_{img}$: Removes the visual loss term from the loss function during pre-training.
- w/o $\mathcal{L}_{text}$: Removes the text loss term from the loss function.

MCRRec+

- w/o $\mathcal{L}_{BPR}$: Replaces BPR Loss with a matching-based InfoNCE loss, while retaining the text semantic regularization term to ensure fair comparison.
- w/o Hybrid Sampling: Replaces the difficulty-aware hybrid sampling strategy with in-batch random negative sampling.

As shown in Appendix Table 3, removing any first-stage component causes performance degradation across datasets. The TGVIA module outperforms a simple MLP by fusing text-aware, context-aware, and raw image features into richer multi-level representations. Both the visual Loss ($\mathcal{L}_{img}$) and text Loss ($\mathcal{L}_{text}$) are critical: removing the former weakens visual representation, while removing the latter harms generalization. The performance gain stems from the joint contribution of TGVIA, $\mathcal{L}_{img}$, and $\mathcal{L}_{text}$ in extracting and integrating personalized multimodal features.

Ablating any component in MCRRec+ also degrades performance. The Hybrid Sampling strategy enhances cover recommendation by

forcing the model to distinguish between visually similar covers, thereby improving specialization without overfitting. Moreover, BPR Loss outperforms InfoNCE, as its ability to learn difficulty-aware ordinal relationships complements the sampling strategy, whereas InfoNCE focuses on batch-level contrast and struggles with fine-grained ranking. Consequently, MCRRec+ benefits from this synergistic design, achieving stronger recommendation performance.

5.4 Further Analysis

We provide a comprehensive analysis to further validate our methods, with complete details in Appendix A.3. Our investigation examines the impact of the pre-training vocabulary (Appendix A.3.1), the number of images per vocabulary item during pre-training (Appendix A.3.2), the number of prompt iterations during inference (Appendix A.3.3), the model’s few-shot learning capability (Appendix A.3.4), efficiency analysis (Appendix A.3.5), and a qualitative case study (Appendix A.4).

6 Conclusion

In this paper, we introduce the cover recommendation task and propose MCRRec and MCRRec+ based on CLIP for the few-shot multimodal setting. Our methods learn multimodal interest features from user profiles, enabling recommendations without specific interaction histories. We propose Visual-Guided Text-Interest Aggregation (VGTIA) to fuse multi-level visual interest guided by interest text into a compact “interest pseudo-token”. Notably, we incorporate multimodal textual inversion techniques to achieve high-quality visual representations of few-shot covers. Through unsupervised pre-training, the MCRRec method acquires strong generalized recommendation capabilities. To further enhance recommendation performance within a specific interest domain, we propose MCRRec+ based on MCRRec. It fine-tunes visual vectors with difficulty-aware hybrid sampling, strengthening the model’s ability to perceive and distinguish subtle differences between semantically similar covers. For a comprehensive evaluation, we construct three cover recommendation datasets and employ a series of recommendation-relevant metrics. Additionally, we propose two new metrics, MPMS and APMS, to measure the semantic match between recommended covers and user interests. Extensive experiments demonstrate that our methods achieve state-of-the-art performance on core metrics, validating superiority.

In the future, we plan to construct higher-quality datasets based on real-world product candidate covers and incorporate additional modalities. We also aim to explore universal representation methods based on Multimodal Large Language Models (MLLMs), extending this research to recommendation scenarios capable of simultaneously processing multimodal content, including audio and video.

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A Appendix

A.1 Dataset Construction

Directly obtaining real-world data on how different item covers affect user interactions for the same item is challenging. To evaluate item cover recommendation tasks and address the lack of task-specific datasets, we constructed new cover recommendation datasets based on existing item recommendation datasets. Selected datasets must satisfy two key criteria: (1) contain real user *explicit interaction interest labels*; and (2) visual information plays a significant role in user-item interactions. Accordingly, we utilized **Pinterest**[16] and **REASONER**[9] public datasets to construct three benchmarks: **Pinterest Cover-small, PC-S** (Small-scale dataset with single-label users.), **Pinterest Cover-large, PC-L** (Larger-scale dataset with single-label users.), and **REASONER Cover, RC** (Large-scale dataset with multi-label users.)

A.1.1 Data Preprocess. To obtain effective data and prepare for the cover construction, we perform preprocessing on the Pinterest and Reasoner datasets respectively.

Pinterest Cover Dataset Introduction: Pinterest[16] is an image-centric recommendation dataset with real user interaction labels from the Pinterest platform. It features high sparsity and is commonly used for visual recommendation. Users “pin” images to

Table 3: Ablation Study Results on MCRc and MCRc+

Dataset	Ablation	MPMS	APMS	Recall@1
PC-S	w/o TG VIA	0.8143	0.7509	0.7572
	w/o $\mathcal{L}_{img}$	0.8524	0.8058	0.7962
	w/o $\mathcal{L}_{text}$	0.7515	0.6859	0.6871
	MCRc	0.9404	0.8778	0.9143
	w/o Hybrid Sampling	0.9410	0.8758	0.9165
	w/o $\mathcal{L}_{BPR}$	0.9512	0.8857	0.9298
PC-L	MCRc+	0.9517	0.8845	0.9332
	w/o TG VIA	0.6845	0.5117	0.4180
	w/o $\mathcal{L}_{img}$	0.6640	0.5039	0.3748
	w/o $\mathcal{L}_{text}$	0.6808	0.5081	0.4095
	MCRc	0.7266	0.5455	0.4762
	w/o Hybrid sampling	0.7343	0.5498	0.4919
RC	w/o $\mathcal{L}_{BPR}$	0.7330	0.5505	0.4919
	MCRc+	0.7351	0.5509	0.4949
	w/o TG VIA	0.5937	0.3368	0.1656
	w/o $\mathcal{L}_{img}$	0.5698	0.3264	0.0862
	w/o $\mathcal{L}_{text}$	0.5922	0.3359	0.1557
	MCRc	0.5982	0.3386	0.1614
	w/o Hybrid Sampling	0.6158	0.3465	0.1887
	w/o $\mathcal{L}_{BPR}$	0.6122	0.3456	0.1782
	MCRc+	0.6174	0.3467	0.1914

personal “boards” and assign “category” labels, aligning with our requirements for explicit user interest labels and visual importance. Due to a lack of image tags, we assigned the user’s valid interaction interest label to the image.

Pinterest Cover-small Preprocess: To ensure high quality, strict filtering was applied. First, images with fewer than five interactions or download failures were removed. Then, users with invalid images or fewer than five interactions were removed. Finally, non-existent image tags were filtered out using the new user interaction interest labels. Each interaction record includes a user interest label (from the interaction), a target item cover (the interacted-upon image), and an item cover label (the image tag). The statistics of the preprocessed dataset are shown in Table 4.

Pinterest Cover-large Preprocess: We applied an interaction threshold of two for images and five for users. Detailed statistics are provided in Table 4.

REASONER Cover Dataset Preprocess: REASONER[9] is an explainable recommendation dataset featuring multi-aspect real user interest labels, video metadata, and user feedback (e.g., viewing intent, multi-dimensional interests, ratings). It was collected via a custom video recommendation platform that displayed titles, tags, and keyframes. To adapt it for cover recommendation, we performed the following preprocessing: interactions with negative intent (e.g., “don’t want to watch”) and non-semantic tags were removed; the middle frame of a video’s keyframe sequence was used as the visual item cover for its representativeness; the user’s multi-aspect interest labels were retained as the user interest label, and the video tags were used as the item cover label. Due to the high original quality of the dataset, no additional user or image

removal was necessary. The final statistics are presented in Table 4. Compared to Pinterest, this dataset provides multiple labels per user and a significantly larger label space.

Table 4: Dataset Statistics

Dataset	#Users	#Covers	#Interest Labels	#Interactions
PC-S	83	987	25	1757
PC-L	2,090	32,160	396	47,627
RC	2,997	4,484	1,576	9,549

A.1.2 Cover Recommendation Dataset Construction. To simulate varying covers for the same item, we further processed the data:

1. Multimodal Interest Representation: For each interest label, select the top 3 most-interacted images as preferred visual information, forming the multimodal “label-image” representation.

2. Validation/Test Set Splitting: Interactions involving images used for multimodal interest representation were first removed. The remaining interactions were then partitioned by user and interest into validation and test sets with a 1:1 ratio. To ensure the completeness of the test set, interactions where a user had only one item for a specific interest label were exclusively assigned to the test set.

3. Candidate Cover Generation: For each user interaction in the validation/test sets, we used the pre-trained CLIP B/32⁶ to compute the cosine similarity between the target image and all other images. We selected the top-K (K=9) most visually similar images as candidate covers, excluding those sharing the target interest label to ensure visual and semantic diversity.

A.2 Introduction on baseline methods

Detailed descriptions of baselines are provided below:

- CLIP-image only: It uses only the mean CLIP feature of interest images as the interest representation.
- CLIP-text only: It uses only the text feature of the interest label to represent interest.
- CLIP-multimodal: It fuses the textual feature of the interest label and the mean-pooled visual feature of the interest images to get interest representation.
- Freedom[14]: It is a training-free, CLIP-based method that retrieves three proxy images and inverts five discrete words via memory banks, then fuses them through high-frequency filtering to form composite queries.
- OsrCIR[41]: It is a training-free method for compositional image retrieval. It uses a MLLM (Qwen-VL-max) to generate target descriptions via a Reflective Chain-of-Thought, with a fixed manipulation text “related to label”, and retrieves images based on CLIP similarity.
- PALAVRA[10]: This two-stage method employs a Set network for personalized textual inversion, pre-training on general image sets before fine-tuning on specific label-image pairs. We adapted it from CLIP-B/32 to CLIP-L/14 by retraining as per the paper and code.

⁶<https://huggingface.co/openai/clip-vit-base-patch32>

- Pic2Word[38]: It converts image embeddings into pseudo-language tokens via a lightweight network trained on unlabeled data, based on CLIP.
- SEARLE[2]: This method obtains compact image representations in language space via a two-stage process: iterative optimization for pseudo-token generation, followed by knowledge distillation into a lightweight network.
- LinCIR[19]: This method operates exclusively on language data, training a projection module through a self-supervised strategy termed Self-Masking Projection.

A.3 In-Deep Analysis

Table 5: Effect of different training vocabulary dictionaries

Dataset	vocabulary	MPMS	APMS	Recall@1
PC-S	COCO	0.9307	0.8736	0.8998
	Laion bigrams	0.9242	0.8633	0.8931
	Open Images V7	0.9404	0.8778	0.9143
PC-L	COCO	0.7163	0.5373	0.4616
	Laion bigrams	0.7210	0.5415	0.4680
	Open Images V7	0.7266	0.5455	0.4762
RC	COCO	0.6000	0.3408	0.1658
	Laion bigrams	0.6006	0.3392	0.1623
	Open Images V7	0.5982	0.3386	0.1614

A.3.1 Effect of different pre-training vocabulary dictionaries. To analyze how different label vocabularies affect model pre-training, we conduct experiments using three vocabularies: COCO dataset[4] (~12K classes), Laion bigrams dataset[4] (~10K classes), and Open Images V7 dataset⁷ (~20K classes). Table 5 reports MCRec’s performance after pre-training on each vocabulary without downstream fine-tuning. The results show consistently strong performance across all three vocabularies, indicating that the model is robust to changes in the label vocabulary. Specifically, Open Images V7 confers a modest advantage on the PC-S and PC-L benchmarks, while all three vocabularies yield comparable results on RC. Overall, Open Images V7 produces slightly better performance; therefore, we adopt Open Images V7 during pre-training.

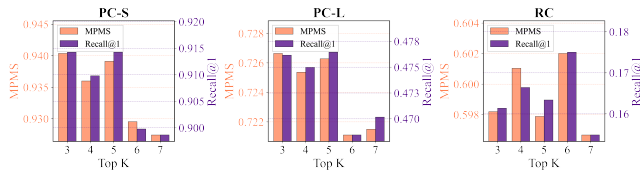


Figure 3: The effect of the top number of images per vocabulary

A.3.2 The impact of the number of images per vocabulary during pre-training. During pre-training, we assign the k most similar images

to each label as its visual representation, where k is a key parameter.

⁷<https://storage.googleapis.com/openimages/web/index.html>

As shown in Figure 3, we evaluate model performance on the three datasets for k values ranging from 2 to 6. The results indicate that overall performance is relatively stable across different k and that the optimal k varies little between datasets. Nevertheless, a certain degree of performance fluctuation can be observed, which can be primarily attributed to two factors: (1) heterogeneous numbers of training images per vocabulary, and (2) varying degrees of intra-label semantic consistency. Intuitively, increasing k enlarges the training set and can improve generalization, but newly included images may be less well aligned with the label and thus introduce noise that degrades performance. Considering the trade-off between training efficiency and model accuracy, we select $k = 2$ as the optimal value. This setting maintains strong performance while limiting training overhead.

Table 6: Effect of indicating the prompt iteration times during inference

Dataset	Iter.	MPMS	APMS	Recall@1
PC-S	0	0.9381	0.8753	0.9120
	1	0.9404	0.8778	0.9143
	2	0.9404	0.8778	0.9143
	3	0.9404	0.8778	0.9143
PC-L	0	0.7265	0.5453	0.4756
	1	0.7266	0.5455	0.4762
	2	0.7267	0.5457	0.4761
	3	0.7266	0.5456	0.4761
RC	0	0.5978	0.3385	0.1597
	1	0.5982	0.3386	0.1614
	2	0.5977	0.3387	0.1609
	3	0.5979	0.3387	0.1612

A.3.3 Influence of Prompt Iteration Times during Inference. To investigate the impact of prompt iterative optimization on model feature extraction capabilities during inference, we conduct comparative experiments on prompt iteration times. Specifically, we set the number of prompt iterations to 0–3 during inference and evaluated model performance across key metrics on three datasets, with results presented in Table 6. Compared with using the original prompt (0 iterations), applying iterative optimization yields measurable improvements. This indicates that the model possesses semantically aware feature extraction capabilities, and prompt optimization enhances the extraction of shared features. Further analysis reveals rapid convergence in the refinement process, with performance gains plateauing after the first iteration, which suggests efficient semantic calibration by the model. Notably, on the RC, a single iteration significantly improved Recall@1 by 1.06%. Balancing inference efficiency and accuracy, we adopt one prompt iteration as the default inference strategy, since it provides significant performance gains while containing computational cost.

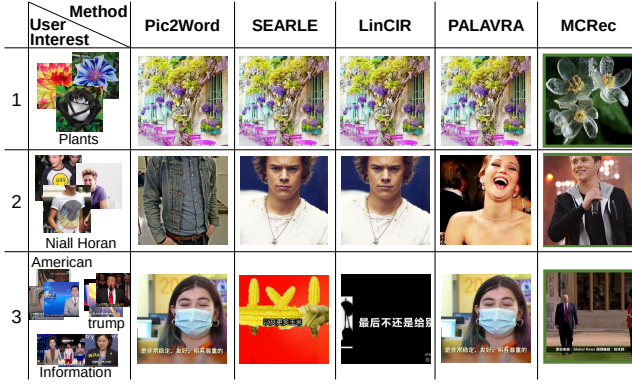


Figure 5: Cover recommendation cases on three datasets

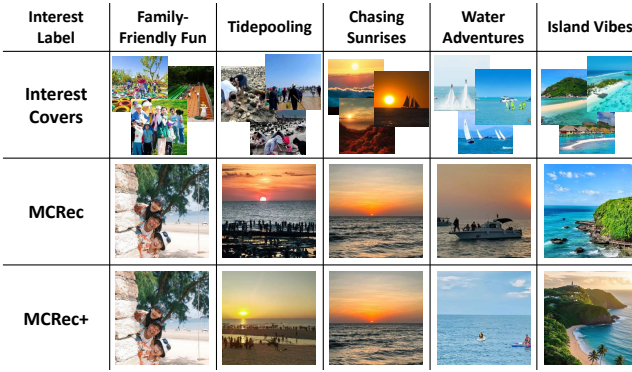


Figure 6: Cover recommendation case in Weizhou Island

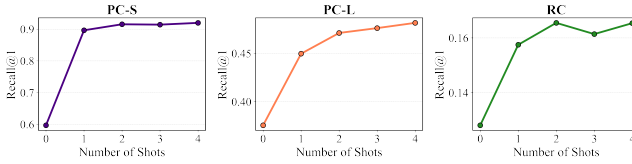


Figure 4: Influence of the shot number used to personalize

A.3.4 Few-shot learning capability. To evaluate the model’s capacity in few-shot learning, we design experiments that control the number of preference images per interest label. Specifically, for each interest label, we use $k = 0, 1, 2, 3, 4$ shots as interest cover set inputs to measure performance under varying shots. Figure 4 illustrates the trend of the model’s Recall@1 across different datasets as k changes. In the zero-shot scenario ($k = 0$), we adapt the model by retrieving the three most similar images from the pre-training image corpus using the label text. Experimental results indicate poor performance in this setting, primarily because retrieved images lack personalized expression and struggle to accurately capture the visual interest. When $k = 1$, precision improves substantially because the model receives a personalized interest image whose distribution matches the target, allowing it to capture user preferences more accurately. As k increases further, performance continues to improve

but exhibits diminishing marginal returns. When $k = 3$, the model achieves optimal and stable performance levels. In summary, the model demonstrates effective learning capabilities in both zero-shot ($k = 0$) and few-shot ($k = 1 \sim 4$) settings, validating its adaptability and robustness under few-shot conditions.

Table 7: Comparison of average inference time (seconds) per interest (Results report average over 10 runs).

Method	PC-S	PC-L	RC	Avg.
PALAVRA	1.290 ± 0.009	1.513 ± 0.005	0.757 ± 0.005	1.187
MCRc	0.042 ± 0.001	0.049 ± 0.001	0.0253 ± 0.001	0.014
MCRc+	0.827 ± 0.005	1.220 ± 0.003	0.428 ± 0.001	0.683

A.3.5 Efficiency analysis. We compare the inference time per interest of our proposed MCRc and MCRc+ against the fine-tuning method PALAVRA. As shown in Table 7, averaged over 10 runs, MCRc’s runtime is approximately two orders of magnitude faster than PALAVRA. MCRc+ offers a further performance gain, being an additional 30% faster than MCRc. This demonstrates a significant advantage in inference efficiency. Furthermore, our method is highly data-efficient. MCRc was pre-trained on only $\sim 10k$ images and $\sim 40k$ label sentences. This represents only 0.4%, 9.8%, and 0.8% of the data required by training-free methods Pic2Word, SEARLE, and LinCIR, respectively. Thus, our approach achieves efficient inference while drastically reducing training resource consumption.

A.4 Case Study

We select one real user–item cover interaction from each of the three constructed datasets to illustrate recommendation results, shown in Figure 5. From top to bottom, the figure shows one instance from PC-S, PC-L, and RC, with correct recommendations outlined in green. MCRc reliably retrieves covers that match actual user interactions, while competing methods sometimes miss visually salient cues. For example, Case 1’s recommendation contains “flowers” but also irrelevant background elements compared with the interest images; Case 2 shows failures to capture sufficiently fine-grained visual details; and in Case 3, faced with images that differ greatly, they struggle to capture the core semantic content. These examples highlight MCRc’s advantage in aligning recommendations with users’ visual preferences.

We further validate our approach with a case study on Weizhou Island attractions. We crawl roughly 120 images for five interest labels: Family-Friendly Fun, Tidepooling, Chasing Sunrises, Water Adventures, and Island Vibes. Using these data, we run cover recommendation with MCRc and MCRc+. As shown in Figure 6, MCRc’s recommendations are visually and semantically consistent with the target interest labels for all categories except Water Adventures. MCRc+ further improves results. It recommends surfing scenes for Water Adventures and produces more accurate island panoramas for Island Vibes. This case study demonstrates that MCRc effectively supports multi-interest cover recommendation and that MCRc+ further enhances recommendation quality.